Definite Integration

Exam code: 9MA0

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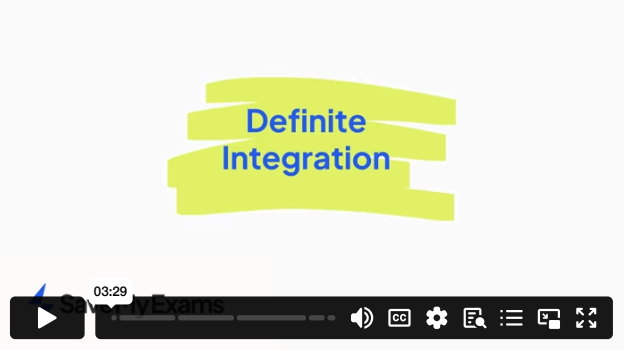

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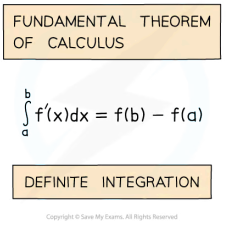
Help me understand

Definite integration

What is definite integration?

- **Definite Integration** occurs in an alternative version of the Fundamental Theorem of Calculus
- This version of the Theorem is the one referred to by most AS/A level textbooks/websites



- **a** and **b** are called limits
  - **a** is the lower limit
  - **b** is the upper limit
- **f'(x)** is the **derivative** of **f(x)**

Why is there no constant of integration in definite integration?

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

$f'(x) = 9x^2$

$\therefore\ f(x) = 3x^3 + c$ INDEFINITE INTEGRATION

$f(2) = 3(2)^3 + c = 24 + c$

$f(5) = 3(5)^3 + c = 375 + c$

$\therefore\ f(5) - f(2) = (375 + c) - (24 + c)$

$= 375 - 24 + c - c$

$= 351$

DEFINITE INTEGRATION

c's CANCEL

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- "**+c**" would appear in both **f(a)** and **f(b)**
  - Since we then calculate **f(b) – f(a)** they cancel each other out
  - There would be a "**+c**" from **f(b)** and a **–**"**+c**" from **f(a)**
- So "**+c**" is **not included** with definite integration

How do I find the value of a definite integral?

- STEP 1: If not given a name, call the integral
  - This saves you having to rewrite the whole integral every time!
- STEP 2: If necessary rewrite the integral into a more easily integrable form
  - Not all functions can be integrated directly
- STEP 3: Integrate without applying the limits
  - Notation: use square brackets [ ] with limits placed after the end bracket
- STEP 4: Substitute the limits into the function and calculate the answer

e.g. EVALUATE $\int_1^9 \left(2x^2 + \frac{3}{x^2} - 4\sqrt{x}\right) dx$

STEP 1 NAME THE INTEGRAL, I

LET $I = \int_1^9 \left(2x^2 + \frac{3}{x^2} - 4\sqrt{x}\right) dx$

STEP 2 REWRITE IN A MORE EASILY INTEGRABLE FORM

$I = \int_1^9 \left(2x^2 + 3x^{-2} - 4x^{\frac{1}{2}}\right) dx$

STEP 3 INTEGRATE

$I = \left[\frac{2}{3}x^3 - 3x^{-1} - \frac{8}{3}x^{\frac{3}{2}}\right]_1^9$

- USE SQUARE BRACKETS
- LIMITS AT END

STEP 4 SUBSTITUTE LIMITS AND CALCULATE

$I = \left(486 - \frac{1}{3} - 72\right) - \left(\frac{2}{3} - 3 - \frac{8}{3}\right)$

$I = \frac{1256}{3}$

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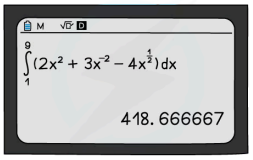
How do I find a definite integral on my calculator?



- (Note how the calculator did not return the exact value $\left(\frac{1256}{3}\right)$ of the integral)

Examiner Tips and Tricks

- Look out for questions that ask you to find an **indefinite** integral in one part (so "**+c**" needed), then in a later part use the same integral as a **definite** integral (where "**+c**" is not needed).

Worked Example

Find the value of

$$\int_2^4 3x(x^2 - 2)\,dx$$

Start by expanding the brackets inside the integral

$$\int_2^4 (3x^3 - 6x)\,dx$$

Integrate as usual (here it's a 'powers of x' integration)

Write the answer in square brackets with the integration limits outside

$$\int_2^4 (3x^3 - 6x)\,dx = \left[3\left(\frac{x^{3+1}}{3+1}\right) - 6\left(\frac{x^{1+1}}{1+1}\right)\right]_2^4$$
$$= \left[\frac{3}{4}x^4 - 3x^2\right]_2^4$$

Now substitute 4 into that function
And subtract from it the function with 2 substituted in

$= 144 - 0$
$= 144$

$$\int_2^4 3x(x^2 - 2)\, dx = 144$$

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